

Extractive continuous process gas analysis

Series 6

FIDAMAT 6

General information

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Overview



The FIDAMAT 6 gas analyzer is suitable for the determination of the total hydrocarbon content in the air and high-boiling gas mixtures.

Benefits

The FIDAMAT 6 gas analyzer is distinguished by its wide range of applications:

- In the presence of up to 100 % H₂O vapor
- In ultra-pure gas applications
- With high-boiling components (up to 200 °C)
- In the presence of corrosive gases (with preliminary filter)

The FIDAMAT 6 exhibits:

- Extremely low cross-sensitivity to interfering gases
- Low consumption of combustion air
- Low influence of oxygen on measured value

The analyzer is additionally equipped with warning and fault messages:

- For failure of combustion gas
- If the flame is extinguished
- To indicate pump and filter faults

Application

Areas of application

- Environmental protection
- Wastewater (in conjunction with a stripping device, verification of the hydrocarbon content of liquids)
- TLV (Threshold Limit Value) monitoring at places of work
- Quality monitoring
- Process exhaust monitoring
- Ultra-pure gas measurements in media such as O₂, CO₂, inert gases and cold sample gases
- Measurement of corrosive and condensing gases
- Process optimization

Further applications

- Chemical plants
- Gas manufacturers (ultra-pure gas monitoring)
- Research and development
- Cement industry (measurement of emissions)
- Paint shops and dry-cleaning systems
- Refineries (tank farms, wastewater)
- Drying systems
- Solvent recovery systems
- Pharmaceutical industry
- Automotive industry (engine development, engine and transmission development and certification)

Special applications

Special applications

Special applications are available on request in addition to the standard combinations, e.g. measuring range 0 to 100 %.

Performance-tested version

Configuration prepared based on QAL1 according to EN 15267 for systems 13th BlmSchV/27th BlmSchV and TA Luft.

Design

- 19" slide-in module with 4 HU for installation
 - In hinged frame
 - In cabinets with or without telescopic rails
- Front plate can be swung down for servicing purposes (laptop connection)
- Gas connections for sample gas inlet and outlet as well as combustion gas and combustion air; pipe diameter 6 mm or 1/4"
- Gas and electrical connections at the rear
- Internal gas paths: stainless steel (mat. no. 1.4571)

Display and control panel

- Large LCD field for simultaneous display of
 - Measured value
 - Status bar
 - Measuring ranges
- Contrast of LCD panel adjustable using menu
- Permanent LED backlighting
- Washable membrane keyboard with five softkeys
- Menu-driven operation for parameterization, test functions, adjustment
- User help in plain text
- Graphic display of concentration trend; programmable time intervals

Input and outputs

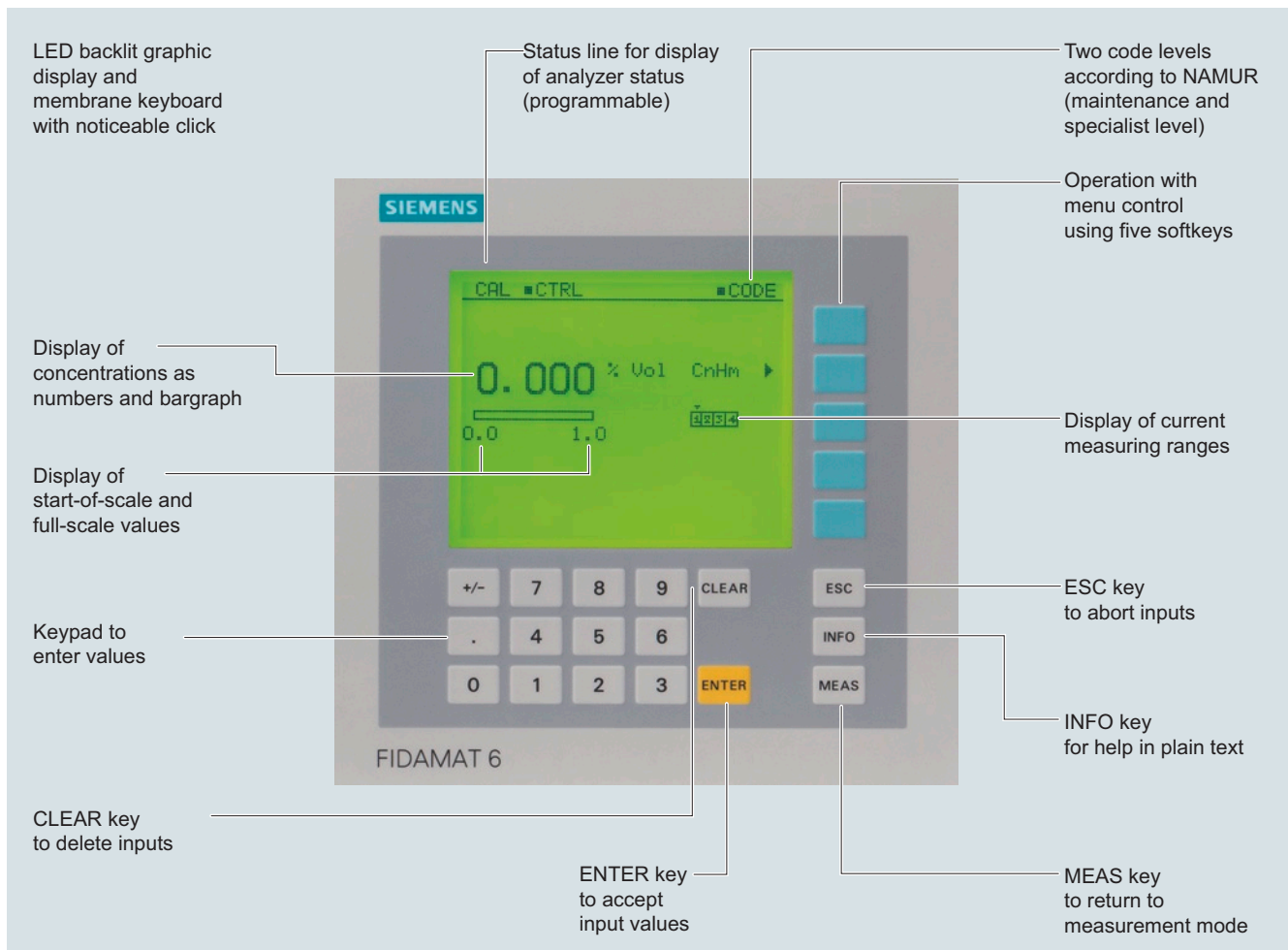
- One analog output for each measured component
- Two programmable analog inputs
- Six digital inputs freely configurable (e.g. for measurement range switchover, processing of external signals from sample preparation)
- Six relay outputs freely configurable (failure, maintenance demanded, maintenance switch, limit alarm, external solenoid valves, measuring point switchover)
- Expansion by eight additional digital inputs and eight additional relay outputs for autocalibration with up to four calibration gases

Communication

RS 485 present in basic unit (connection from the rear).

Options

- RS 485/RS 232 converter
- RS 485/Ethernet converter
- RS 485/USB converter
- Incorporation in networks via PROFIBUS DP/PA interface
- SIPROM GA software as service and maintenance tool



FIDAMAT 6, membrane keyboard and graphic display

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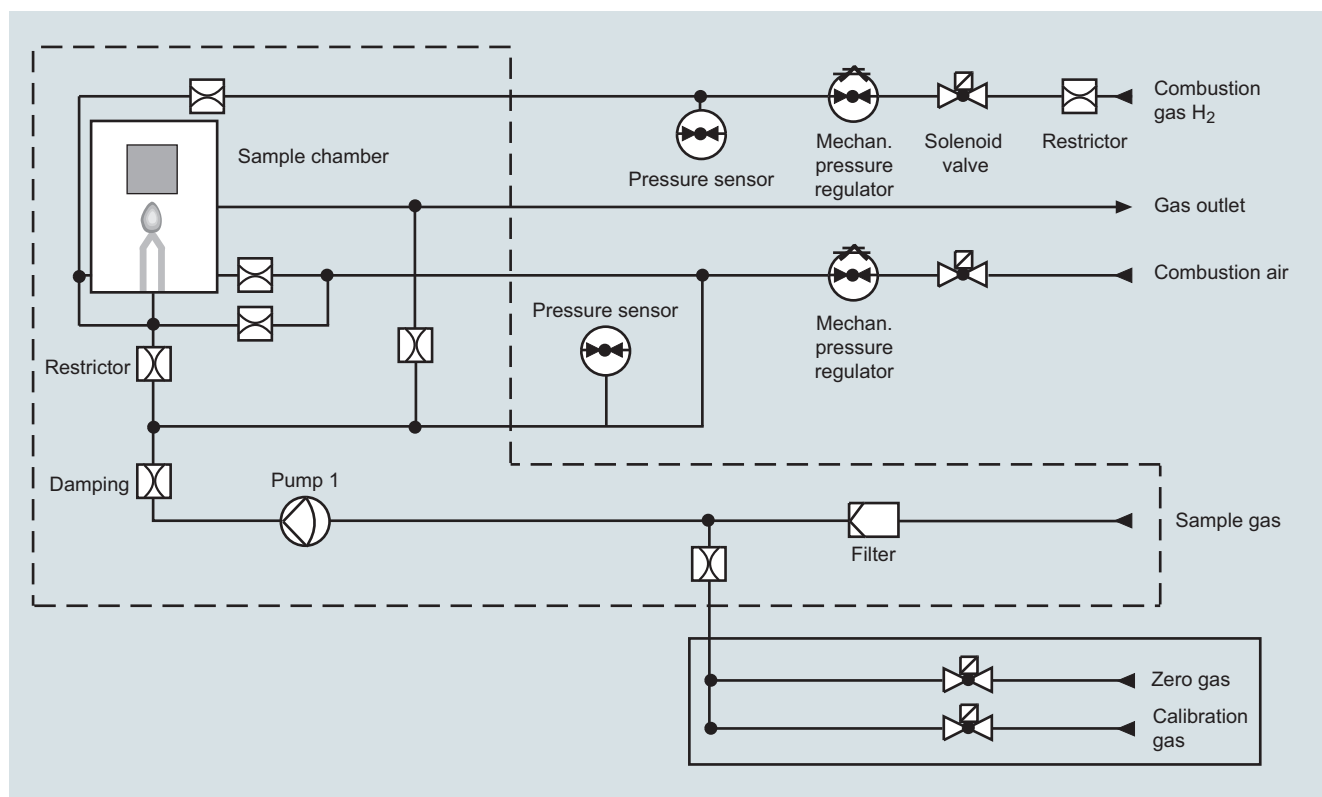
FIDAMAT 6

General information

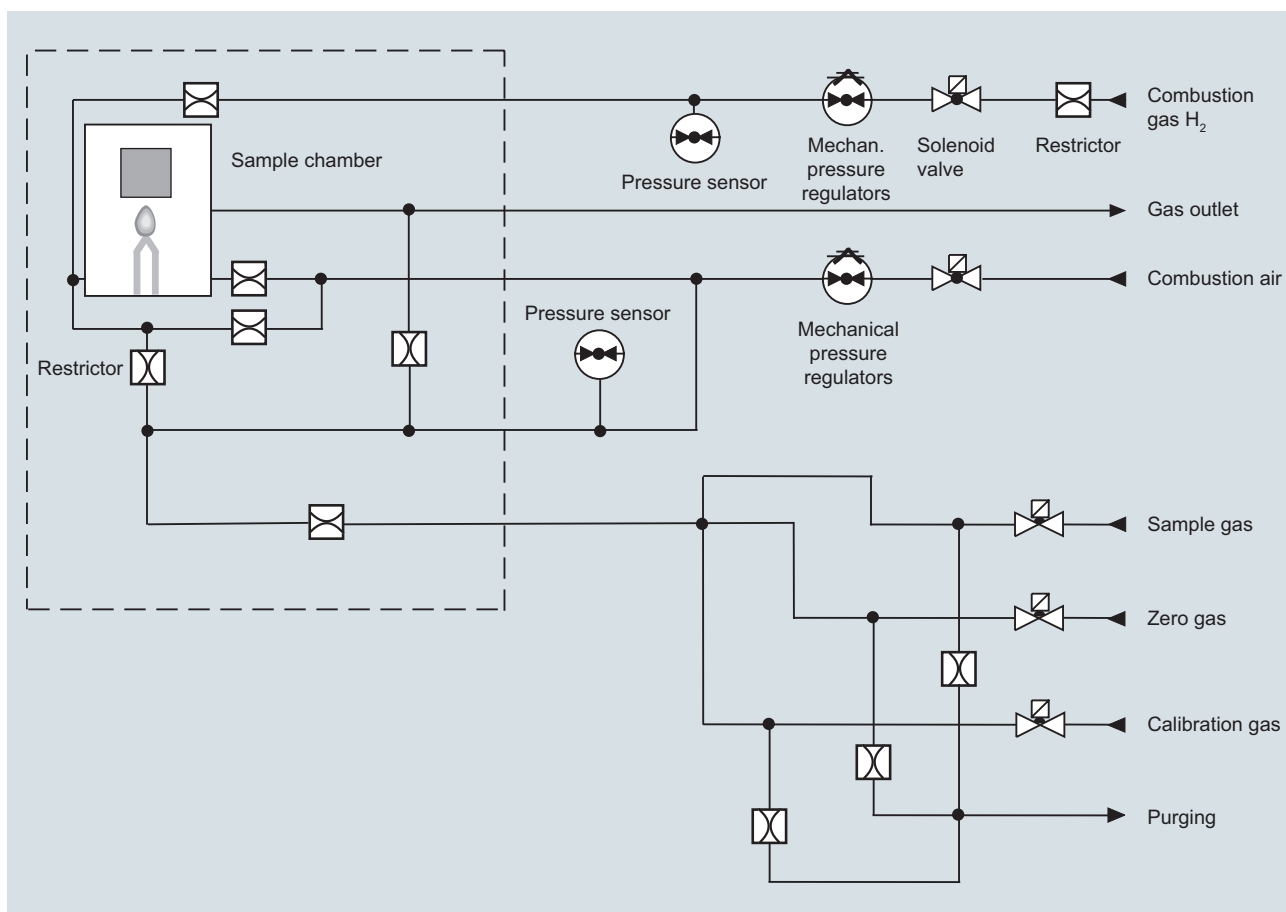
Designs – parts wetted by sample gas

Gas path	Material
Piping	Stainless steel, mat. no. 1.4571
Gas inlet	Stainless steel, mat. no. 1.4571
Gaskets	Graphite
Sample gas restrictor	Quartz
Auxiliary gas restrictors	Stainless steel, mat. no. 1.4571
Pump membrane	PTFE
Pump head	Stainless steel, mat. no. 1.4571
Detector	
• Nozzle	Quartz
• FID housing	Stainless steel, mat. no. 1.4571

Gas path



FIDAMAT 6 total hydrocarbon analyzer, gas path with pump and with connection for combustion air



FIDAMAT 6 total hydrocarbon analyzer, gas path without pump and with connection for combustion air

Extractive continuous process gas analysis

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FIDAMAT 6

General information

Function

Principle of operation

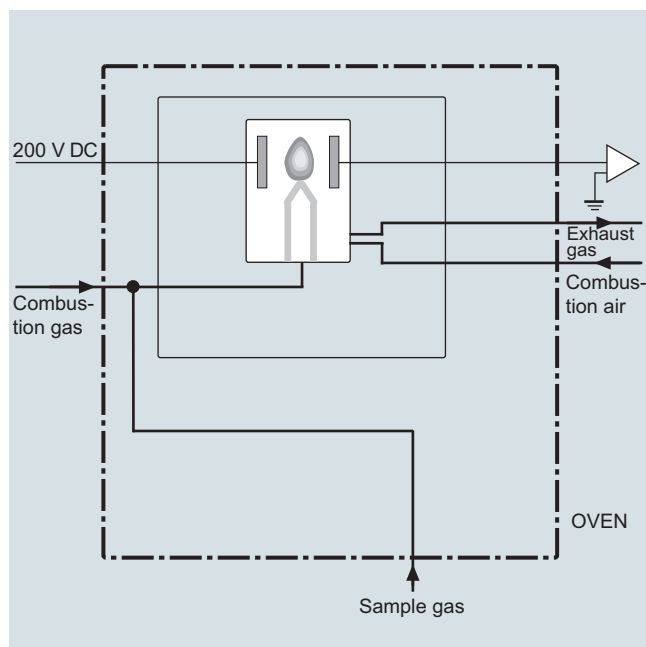
The FIDAMAT 6 carries out substance-specific measurements and not component-specific measurements. It measures the total of all hydrocarbons in a sample gas, but with different weighting of the hydrocarbon molecules. To a first approximation, the display is proportional to the number of C atoms in the respective molecule. However, there are fluctuations in practice. The display deviation for the respective molecule is expressed by the response factor.

The sample gas is supplied to the FIDAMAT 6 through overpressure or drawn in by the built-in diaphragm pump (optionally via a heated line and an additional filter) and passed on to the flame ionization detector via an obstruction-proof fused-silica restrictor.

In the detector, the hydrocarbons in the sample gas are burned in an oxyhydrogen gas flame. Burning partially ionizes the proportion of organically-bound hydrocarbons. The released ions are converted into an ionic current by the voltage present between two electrodes, and measured using a highly sensitive amplifier. The current measured is proportional to the quantity of organically-bound C atoms in the sample gas.

A pressure regulator keeps the combustion gas pressure constant. The balanced system of pump, capillary tubes, and pressure regulator for combustion air ensures that the sample gas pressure is kept constant.

When the analyzer is switched on, ignition is carried out automatically when the setpoint temperature has been reached and, for versions "with pump", the pump is also started up.



FIDAMAT 6, principle of operation

The FIDAMAT 6 provides various messages in the form of floating contacts:

- Maintenance request
E.g. sample gas flow (filter/pump)
Fan failure (advance warning for measuring accuracy)
The measured value remains unaffected.
- Fault
e. g., hydrogen, combustion air and sample gas pressures, temperature, analyzer part and pump, fault in the electronics (temperature).
The measured value may be influenced.
- Failure
In the event of failure of, for example, the electronics, power supply, combustion gas, combustion air or sample gas, the analyzer automatically shuts down (the combustion gas valve is closed).

Note

The sample gases must be fed into the analyzers free of dust. Condensation should be avoided. Therefore, the use of gas modified for the measuring task is necessary in most application cases.

Calibration

The calibrating interval should be adapted to the respective measuring task. We recommend N_2 as zero gas (at least 5.0; for measuring of hydrocarbons < 1 vpm: at least 6.0).

The calibration gas should have a concentration of at least 60 % of the leading measuring range. The concentration of residual hydrocarbons must not exceed 0.1 vpm.

For pure gas measurement, use suitable additional gases.

Example:

1. Emission measurement

Measuring range	0 ... 50 mg C/m ³
Zero gas	N ₂ (5.0 or better)
Calibration gas	21 vpm C ₃ H ₈ in N ₂ (corresponds to 31.43 mg C/m ³ at 20 °C)

2. Purity measurement in 100 % O₂

Measuring range	0 ... 50 vpm C ₁
Zero gas	N ₂ (5.0 or better)
Calibration gas	At least 30 vpm CH ₄ in O ₂

Essential characteristics

- Four measuring ranges which can be freely configured, even with suppressed zero point, all measuring ranges are linear
- Galvanically isolated measured-value output 0/2/4 to 20 mA (also inverted)
- Autoranging possible; remote switching is also possible
- Storage of measured values possible during adjustments
- Measuring range identification
- Measuring point switchover for up to 6 measuring points
- Measuring point identification
- Wide range of selectable time constants (static/dynamic noise suppression); i.e. the response time of the device can be adapted to the respective measuring task
- Easy handling thanks to menu-driven operation
- Low long-term drift
- Two control levels with their own authorization codes for the prevention of accidental and unauthorized operator interventions
- Automatic measuring range calibration can be configured
- Operation based on the NAMUR recommendation
- Customer-specific analyzer options such as:
 - Customer acceptance
 - TAG labels
 - Drift recording
- Wear-free, corrosion-proof filter housing
- No blocking of the sample gas capillaries through the use of a quartz restrictor
- Purge function in the event of analyzer or power supply failure (avoids build-up of toxic and corrosive substances in the device)
- Low consumption of combustion air
- Response factors comply with the minimum requirements in accordance with German air purity guidelines and the Working Group of the German automotive Industry
- Simple handling using a numerical membrane keyboard and operator prompting

Response factors (examples, mean values)

Substance	Mean response factor
n-butane	1.00
n-propane	1.00
n-heptane	1.00
Cyclohexane	1.08
Isopropanol	0.81
Toluene	1.06
Acetone	0.92
Ethyl acetate	0.76
Isobutyl acetate	0.83
Methane	1.06
Ethane	0.99
n-hexane	1.01
iso-octane	1.04
Ethine (acetylene)	0.91
Propene	0.84
Methanol	0.87
Ethanol	0.83
Ethanoic acid	1.13
Methyl acetate	0.67
Benzene	1.01
Ethyl benzene	0.96
p-xylene	1.03
Dichloromethane	1.13
Trichloroethene	1.01
Tetrachlorethene	1.07
Chloroform	0.72
Chlorobenzene	1.15

Cross-interferences (examples)¹⁾

Interfering component	Concentration of the interfering component	Induced cross-interference
O ₂ in N ₂	(21 vol. %)	< 0.3 mg/m ³
SO ₂ in N ₂	(258 mg/m ³)	< 0.15 mg/m ³
NO in N ₂	(310 mg/m ³)	< 0.5 mg/m ³
NO ₂ in synth. Air	(146 mg/m ³)	< 0.1 mg/m ³
CO in N ₂	(461 mg/m ³)	< 0.15 mg/m ³
CO ₂ in N ₂	(18 vol. %)	< 0.1 mg/m ³
HCl in N ₂	(78 mg/m ³)	< 0.3 mg/m ³

¹⁾ With measuring range 0 to 15 mg/m³.

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FIDAMAT 6

19" rack unit

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Technical specifications

General information		Measuring response	
Measuring ranges	4, internally and externally switchable; manual and autoranging possible	Output signal fluctuation	Based on sample gas pressure 1 013 hPa absolute, 0.5 l/min sample gas flow and 25 °C ambient temperature < 0.75% of the smallest possible measuring range according to rating plate, with electronic damping constant of 1 s (corresponds to $\pm 0.25\%$ at 2σ)
Smallest possible measuring span	0 ... 10 vpm	Zero point drift	< 0.5%/month of the smallest possible measuring span according to rating plate
Largest possible measuring span	99.999 vpm ^{*)}	Measured-value drift	< 1%/week of the current measuring range
Concentration units	vpm, C ₁ , C ₃ , C ₆ or mgC/m ³	Repeatability	< 1% of the current measuring range
Autoranging	Hysteresis, selectable	Detection limit	0.1 vpm (version for ultra-pure gas measurement: 50 ppb)
Measured-value display	Digital concentration display (5 digits with floating point)	Linearity error	< 1% of the current measuring range
Resolution of digital display	0.1% of measured value	Influencing variables	
Operating position	Front wall, vertical	Ambient temperature	Based on sample gas pressure 1 013 hPa absolute, 0.5 l/min sample gas flow and 25 °C ambient temperature < 1%/10 K referred to smallest possible span according to rating plate
Conformity	CE mark in accordance with EN 50081-1, EN 50082-2	Atmospheric pressure	< 1%/50 hPa
Oven temperature	Adjustable, 100 ... 200 °C	Sample gas pressure	< 2% of the current measuring range/ 1% pressure change (within 600 ... 1 100 hPa)
Design, enclosure		Auxiliary power	< 1% of the current measuring range with rated voltage $\pm 10\%$
Degree of protection	IP20 according to EN 60529	Position influence	< 1% with < 15° inclination
Weight	Approx. 23 kg	Electrical inputs and outputs	
Electrical characteristics		Analog output	0/2/4 ... 20 mA, isolated; max. load 750 Ω
Auxiliary power	100 ... 120 V AC (nominal range of use 90 ... 132 V), 48 ... 63 Hz or 200 ... 240 V AC (nominal range of use 180 ... 264 V), 48 ... 63 Hz	Relay outputs	6, with changeover contacts, freely configurable, e.g. for measuring range identification; load: 24 V AC/DC/1 A, potential-free
Power consumption	• Approx. 150 VA during operation, • Approx. 350 VA during warm-up phase	Analog inputs	2, dimensioned for 0/2/4 to 20 mA for external pressure sensor and correction of influence of accompanying gas (correction of cross-interference)
EMC interference immunity (electromagnetic compatibility)	In accordance with standard requirements of NAMUR NE21 (08/98)	Digital inputs	6, designed for 24 V, floating, freely configurable, e.g. for measuring range switchover
Electrical safety	In accordance with EN 61010-1, overvoltage category II	Serial interface	RS 485
Fuse values	• 100 ... 120 V: 4.0T/250 • 200 ... 240 V: 2.5 T/250	Options	AUTOCAL function each with 8 additional digital inputs and relay outputs, also with PROFIBUS PA or PROFIBUS DP
Gas inlet conditions		Climatic conditions	
Permissible sample gas pressure		Permissible ambient temperature	5 ... 45 °C in operation, -30 ... +70 °C during storage and transportation
• Without pump • With integrated pump	< 2 000 hPa abs. 600 ... 1 100 hPa	Permissible humidity	< 90% RH (RH: relative humidity) as annual average, during storage and transportation (must not fall below dew point)
Sample gas temperature	0 ... 200 °C		
Sample gas humidity	< 90% RH (RH: relative humidity)		
Dynamic response			
Warm-up period	At room temperature, approx. 2 ... 3 h		
Delayed display (T ₉₀)	2 ... 3 s		
Damping (electrical time constant)	0 ... 100 s, configurable		
Dead time (purging time of the gas path in the unit at 1 l/min)	With filter, 2 ... 3 s		
Time for device-internal signal processing	< 1 s		

*) 100% as special application

FIDAMAT 6 with pump and heated oven, with combustion air connection

Gases	Inlet pressure hPa (abs.)	Operating pressure Pump startup		Flow through FID ml/min	Flow through bypass ml/min
		Without hPa (abs.)	With hPa (abs.)		
Combustion gas	3 000 ... 5 000	2 000 ± 20		~ 25	—
Combustion air	3 000 ... 5 000	1 420 ± 20	1 500	~ 320	~ 500
Sample gas	~ 1000	—	1 500 ± 2	~ 3	~ 1 000
Zero gas	3 500 ... 4 000	—	1 500 ± 2	~ 3	~ 1 000
Calibration gas	3 500 ... 4 000	—	1 500 ± 2	~ 3	~ 1 000

FIDAMAT 6 without pump, with heated oven, with combustion air connection

Gases	Inlet pressure hPa (abs.)	Operating pressure Sample/calibration gas		Flow through FID ml/min	Flow through bypass ml/min
		Without hPa (abs.)	With hPa (abs.)		
Combustion gas	3 000 ... 5 000	2 000 ± 20		~ 25	—
Combustion air	3 000 ... 5 000	1 480 ± 5	—	~ 320	~ 300
Sample gas	1 500 ... 2 000	—	1 500 ± 2	~ 3	~ 500
Zero gas	1 500 ... 2 000	—	1 500 ± 2	~ 3	~ 500
Calibration gas	1 500 ... 2 000	—	1 500 ± 2	~ 3	~ 500

The supply gases (combustion gas, combustion air) must have a degree of purity of 5.0 in order to guarantee correct measurements. The degree of purity must be increased in the case of very small hydrocarbon concentrations (< 1 vpm).

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FIDAMAT 6

19" rack unit

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Selection and ordering data

FIDAMAT 6 gas analyzer

19" rack unit for installation in cabinets

➤ Click on the Article No. for the online configuration in the PIA Life Cycle Portal.

Gas connections

Pipe with 6 mm outer diameter

Pipe with 1/4" outer diameter

Version

Without pump, for sample gas with overpressure¹⁾

With heated pump, for sample gas with atm. pressure

Combustion air feed

With connection for combustion air

Number of channels

1-channel version

Add-on electronics

Without

AUTOCAL function

- With 8 additional digital inputs/outputs
- With 8 digital inputs/8 digital outputs, PROFIBUS PA interface
- With 8 digital inputs/8 digital outputs, PROFIBUS DP interface

Power supply

100 ... 120 V AC, 48 ... 63 Hz

200 ... 240 V AC, 48 ... 63 Hz

Combustion gases

H₂

Language (supplied documentation, software)

German

English

French

Spanish

Italian

Article No.

➤ 7MB2421- - - - A

0

1

B

D

A

1

0

1

6

7

0

1

A

0

1

2

3

4

Additional versions

Order code

Add "-Z" to Article No. and specify Order code

Telescopic rails (2 units)

A31

TAG labels (specific lettering based on customer information)

B03

Clean for O₂ service (specially cleaned gas path)

Y02

Measuring range indication in plain text, if different from the standard setting

Y11

Special setting (only in conjunction with an application No.)

Y12

Extended special setting (only in conjunction with an application No.)

Y13

Configuration according to EN 14181:2004

Y17

Prepared for EN 15267:2015²⁾

Y27

Accessories

Article No.

RS 485/Ethernet converter

A5E00852383

RS 485/RS 232 converter

C79451-Z1589-U1

RS 485/USB converter

A5E00852382

AUTOCAL function each with 8 digital inputs/outputs

C79451-A3480-D511

AUTOCAL function 8 digital inputs/outputs each and PROFIBUS PA

A5E00057307

AUTOCAL function 8 digital inputs/outputs each and PROFIBUS DP

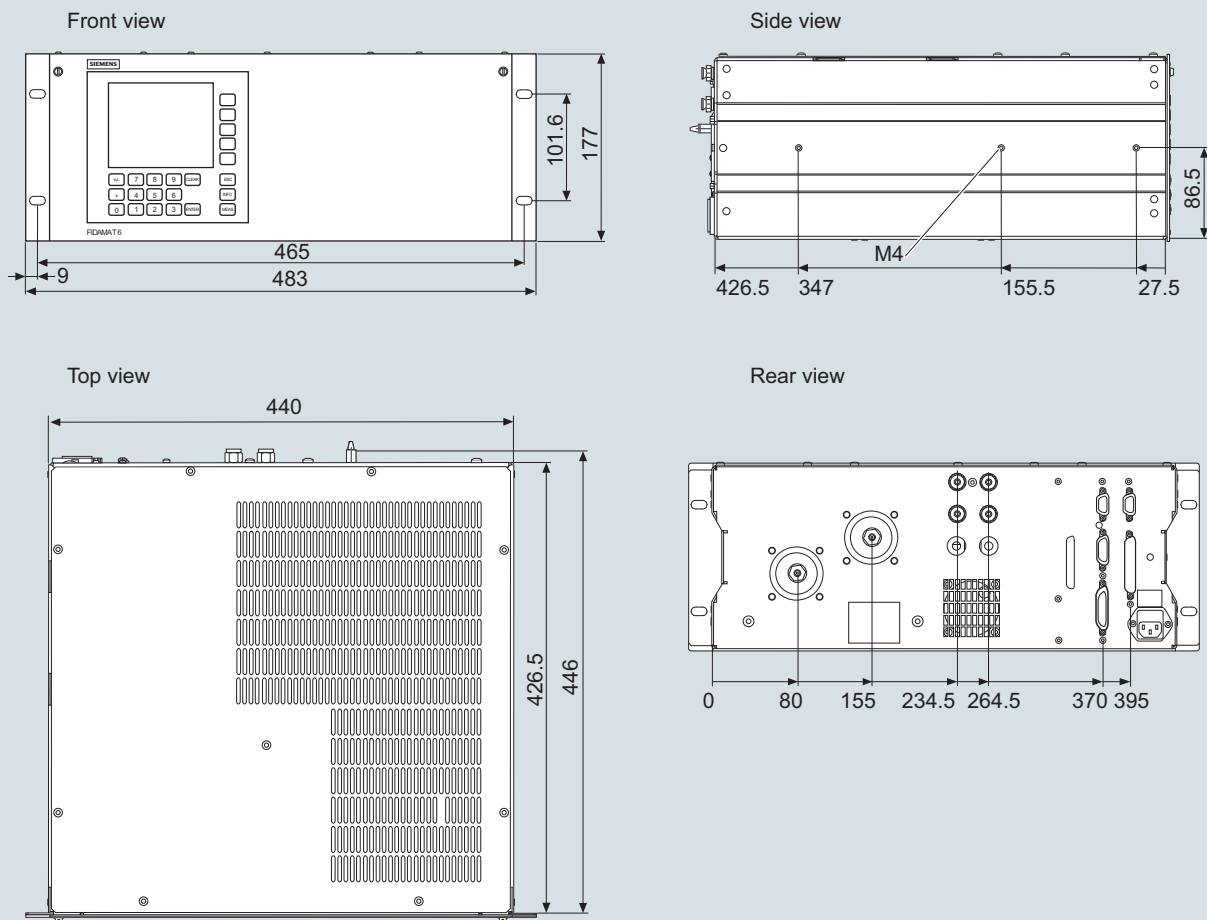
A5E00057312

Set of Torx screwdrivers

A5E34821625

¹⁾ No QAL1 certification according to EN 15267

²⁾ For certified operation, compensation of the cross-interference caused by the oxygen is required. To do this, the device must be supplied with the current oxygen concentration value of the sample gas. The supply takes place over an external measuring instrument that must also meet the requirements of EN 15267-3 (e.g. ULTRAMAT 23 or OXYMAT 6). The FIDAMAT 6E –Y27 is preconfigured accordingly and expects an analog signal of 4 ... 20 mA corresponding to 0 ... 21 vol % O₂ at analog input AI2.

Dimensional drawings

FIDAMAT 6, 19" unit, dimensions in mm

Extractive continuous process gas analysis

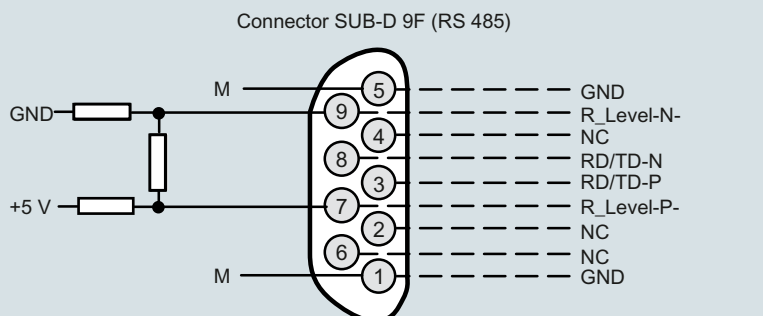
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FIDAMAT 6

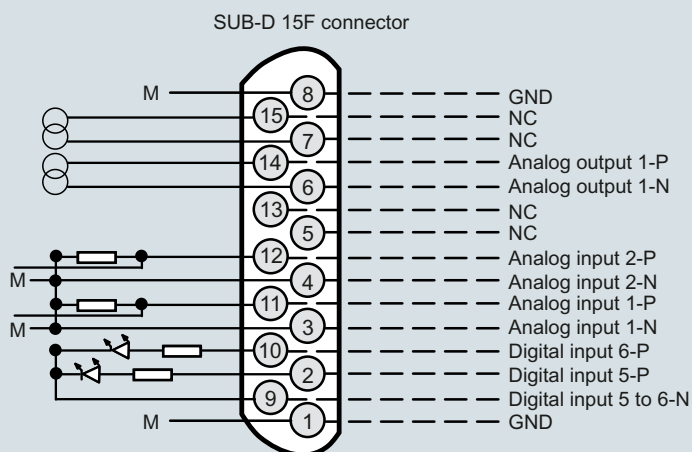
19" rack unit

Circuit diagrams

Pin assignment (electrical and gas connections)



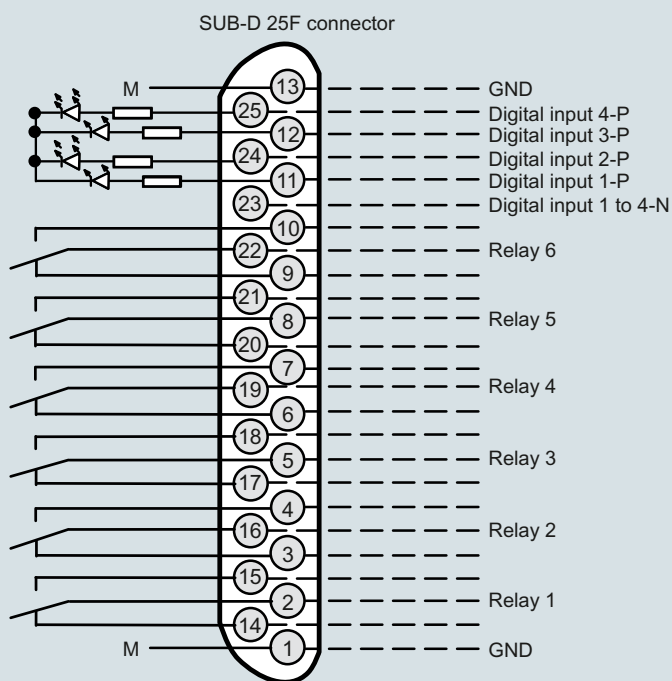
It is possible to connect bus terminating resistors to pins 7 and 9.



Analog outputs isolated (also from each other), $R_L \leq 750 \Omega$

Correction of pressure or cross-interference
Correction of cross-interference
Correction of cross-interference

Analog inputs non-isolated,
0 ... 20 mA/500 Ω
or 0 ... 10 V
(low resistance)



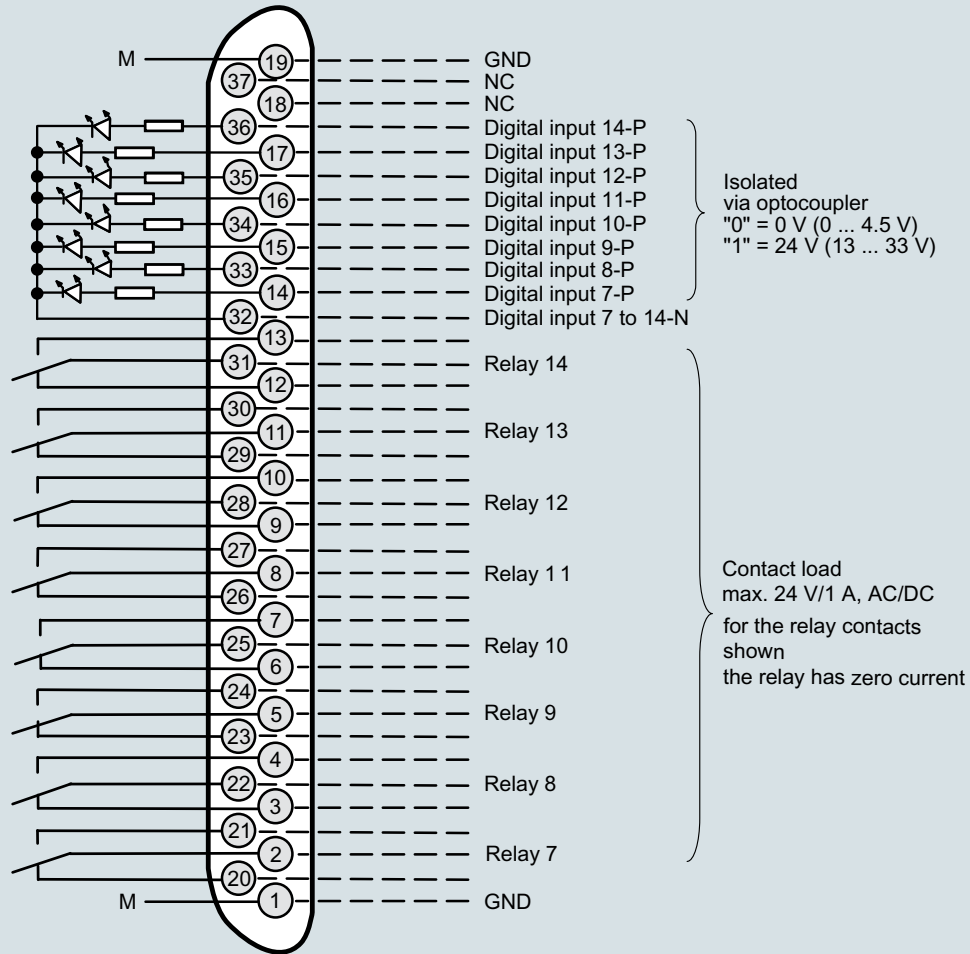
Isolated via optocoupler
"0" = 0 V (0 ... 4.5 V)
"1" = 24 V (13 ... 33 V)

Contact load
max. 24 V/1 A, AC/DC
relay contacts shown:
relay coil has zero current

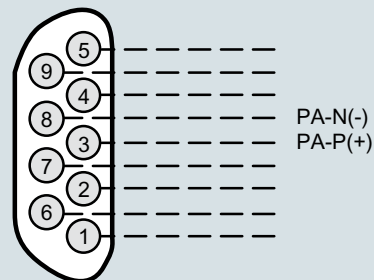
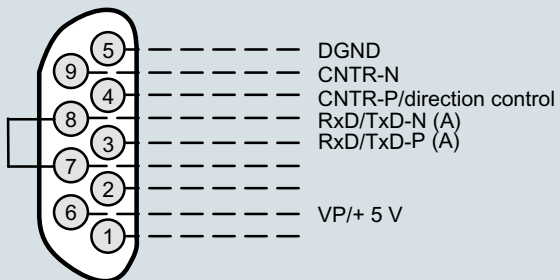
Note:
All cables to the connectors or terminal blocks must be shielded and rest against the enclosure potential.

FIDAMAT 6, 19" rack unit, pin assignment

Connector SUB-D 37F (option)

Connector SUB-D 9F
PROFIBUS DP

optional

Connector SUB-D 9M
PROFIBUS PA**Note:**

All cables to the connectors or terminal blocks must be shielded and rest against the enclosure potential.

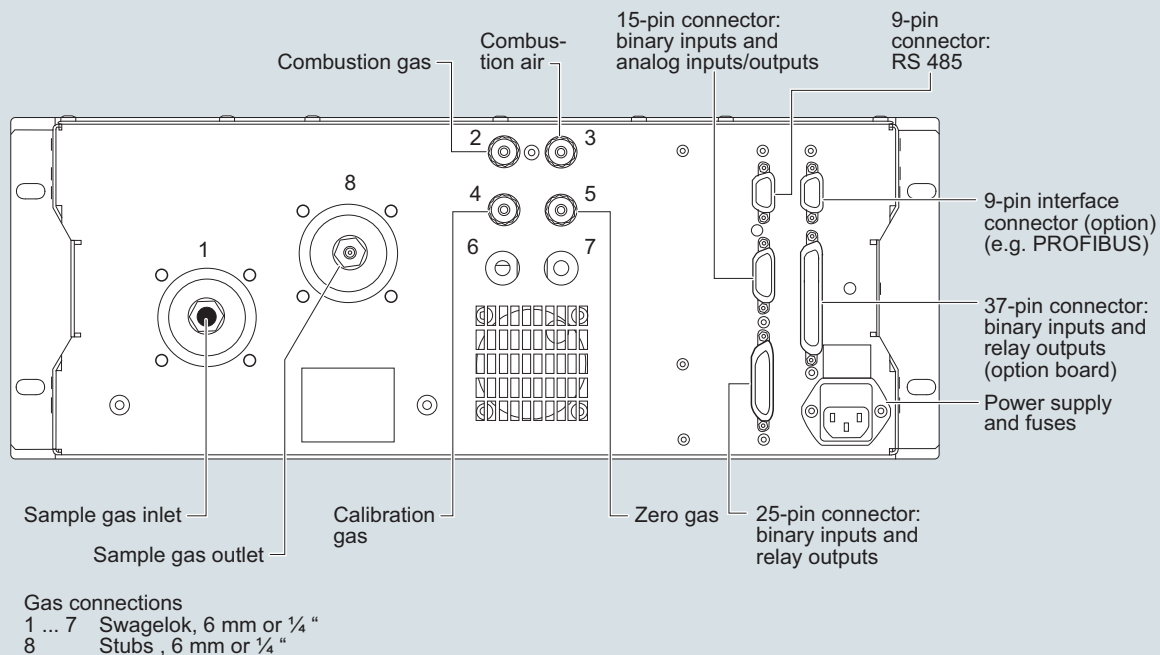
FIDAMAT 6, 19" rack unit, pin assignment of the AUTOCAL board and PROFIBUS connectors

Extractive continuous process gas analysis

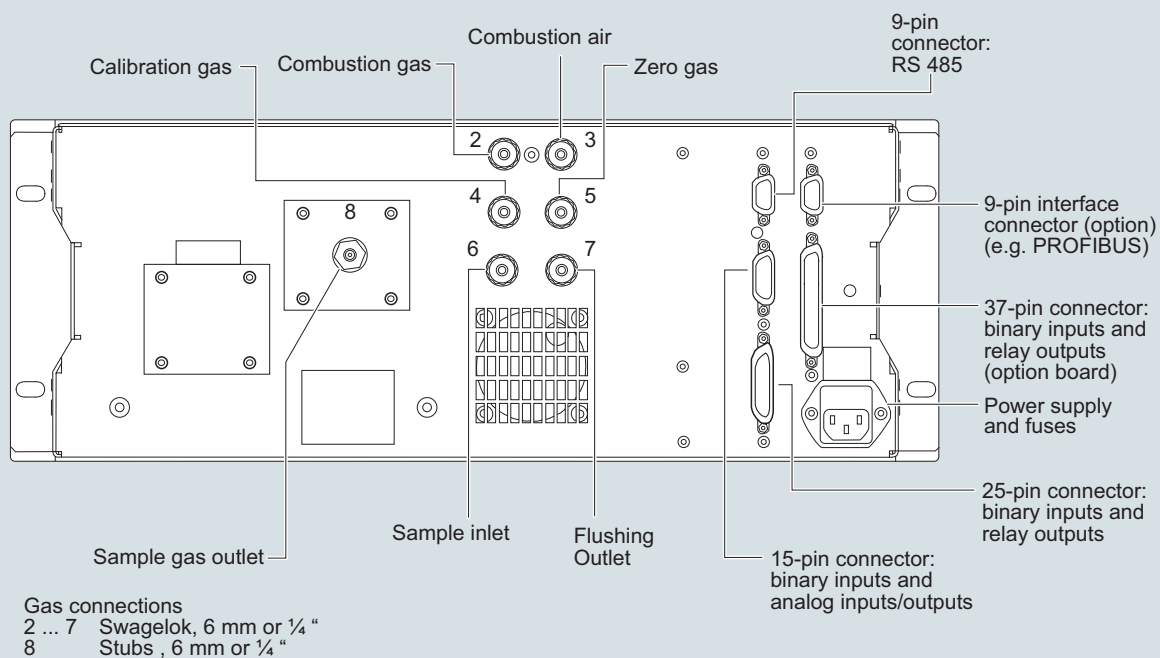
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FIDAMAT 6

19" rack unit



FIDAMAT 6, gas connections and pin assignment, version with pump



FIDAMAT 6, gas connections and pin assignment, version without pump

Selection and ordering data

Operating instructions	Article No.
FIDAMAT 6 Gas analyzer for determination of total hydrocarbon concentration	
• German	A5E00221703
• English	A5E00222135
• French	A5E00222138
• Spanish	A5E00222141
• Italian	A5E00222144
FIDAMAT 6-G Gas analyzer for determination of total hydrocarbon content	
• German	A5E00476038

More information

The complete documentation is available in various languages for downloading free of charge:

<http://www.siemens.com/processanalytics/documentation>

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FIDAMAT 6

Suggestions for spare parts

Selection and ordering data

Description	2 years (quantity)	5 years (quantity)	Article No. FIDAMAT 6	
			With pump	Without pump
Analyzer unit				
FI detector, complete		1	A5E00295816	A5E00295816
Sample gas path				
Pump (KNF)	1	1	A5E00882121	
Set of gaskets for pump (KNF)	4	10	A5E03792459	
Filter, with gasket for sample gas	1	3	A5E00248845	
Pressure regulators	1	1	A5E00248851	A5E00248851
Gasket for pressure regulator	1	2	A5E00295107	A5E00295107
Filter, complete (sample gas inlet, 6 mm)		1	A5E00295928	
Filter, complete (sample gas inlet, 1/4")		1	A5E00295976	
Solenoid valve (1-way)	1	2	A5E00296562	A5E00296562
Solenoid valve (2-way)	1	2	A5E00296565	
Gasket, PTFE, 1.5 mm (20 units)	1	2	C79451-A3040-D101	C79451-A3040-D101
Gasket, graphite, 0.5 to 1 mm (20 units)	1	2	C79451-A3040-D102	C79451-A3040-D102
Gasket, graphite, 1.5 mm (20 units)	1	2	C79451-A3040-D103	C79451-A3040-D103
Gasket, graphite, 3 mm (20 units)	1	2	C79451-A3040-D105	C79451-A3040-D105
Pressure ring, 1 mm (20 units)		1	C79451-A3040-D112	C79451-A3040-D112
Pressure ring, 1.5 mm (20 units)		1	C79451-A3040-D113	C79451-A3040-D113
Pressure ring, 3 mm (20 units)		1	C79451-A3040-D115	C79451-A3040-D115
Outer rings, 0.5 ... 1 mm (20 units)		1	C79451-A3040-D121	C79451-A3040-D121
Outer rings, 1.5 ... 3 mm (1/8") (20 units)		1	C79451-A3040-D122	C79451-A3040-D122
Electronics				
Front plate	1	1	A5E00248790	A5E00248790
Adapter plate	1	1	A5E00248795	A5E00248795
Temperature fuse (retrofitting set)	1	2	A5E01040317	A5E01040317
Fusible element, 230 V AC	2	3	A5E00248819	A5E00248819
Fusible element, 110 V AC	2	3	A5E00248822	A5E00248822
LC display	1	1	A5E00248920	A5E00248920
Cable, temperature sensor for oven		1	A5E00283770	A5E00283770
Cable, temperature sensor for analyzer part		1	A5E00283780	A5E00283780
Cable, magnetic distributor		1	A5E00283800	A5E00283800
Cable, heater for oven, 230 V AC		1	A5E00283817	A5E00283817
Cable, heater for oven, 110 V AC		1	A5E00295469	A5E00295469
Cable, electrode voltage, complete		1	A5E00284092	A5E00284092
Cable, signal cable		1	A5E00284094	A5E00284094
Cable, connecting cable (4-pole)	1	1	A5E00284095	A5E00284095
Cable, connecting cable (5-pole)	1	1	A5E00284096	A5E00284096
Axial-flow fan, 24 V DC		1	A5E00313839	A5E00313839

If the device was supplied with a specially cleaned gas path for high oxygen context ("Clean for O₂ service"), please ensure that you specify this when ordering spare parts. This is the only way to guarantee that the gas path will continue to comply with the special requirements for this version.

SIEMENS General Conditions of SALE of Goods & Services (including Software)

1 Interpretation

1.1 In these Conditions:

"Conditions" means these terms and conditions for the sale and purchase of Supplies which are incorporated into and form part of the Contract.

"Contract" means the contract between the Seller and the Purchaser (subject to the provisions of Clause 2) for the purchase and sale of the Supplies, comprising:-

- (i) the Seller's quotation (including documents (if any) incorporated by express reference on the face of the quotation) and the acceptance thereof by the Purchaser; or
- (ii) the Order and the Seller's acceptance thereof,

together with these Conditions and any written agreement pursuant to Clause 2.2.1.

"Deliver", "Delivered" or "Delivery" means the act of:

- (i) delivering or making available the Supplies ex Works or unless otherwise agreed in writing between Purchaser and Seller; or
- (ii) where assembly, installation, erection or commissioning of the Supplies at Site or otherwise is required, the Seller declaring that the Supplies are ready for acceptance.

"Delivery Date(s)" means the date or dates provided in the Contract, on which Delivery of the Supplies or instalments of the Supplies are to take place respectively; or the extended date or dates for the same pursuant to the notification issued by the Seller in accordance with Clause 6.12.

"ex Works" means that the Supplies are placed at the disposal of the Purchaser at the Seller's premises or another named place, not cleared for export and not loaded on any collecting vehicle.

"Goods" means the goods (including any instalment of the goods or any parts of them) which the Seller is to supply in accordance with the Contract and which may (without limitation) comprise or include Software.

"Order" means the written order placed by the Purchaser for the supply of the Goods and/or Services and documents (if any) incorporated by express reference on the face of the Order.

"Party" means the Seller or the Purchaser and "Parties" means both of them.

"Purchaser" means the person, firm or company who accepts a quotation of the Seller for the sale of the Goods and/or Services or who places the Order for the Goods and/or Services.

"Purchaser's Default" means a default of the Purchaser in the manner described in Clause 6.13.

"Seller" means the Siemens company referred to on the face of the Order with whom the Contract is made by the Purchaser.

"Services" means the provision of work and/or other services which are the subject of the Contract that may (without limitation) comprise and include Software and which may relate to installation in accordance with these Conditions.

"Site" means the address for delivery, that part of the address for delivery where the Goods are to be installed and operated, or where any Services are to be performed.

"Software" means the software and firmware items which are comprised or included in or related to the Supplies.

"Supplies" means Goods and/or Services.

- 1.2 Any reference in the Contract to a statutory provision shall include that provision and any regulations made pursuant thereto which may from time to time be modified or re-enacted in accordance with the laws of Singapore. For the avoidance of doubt, such modifications or re-enactment of statutory provisions shall apply to the Contract prior to completion of the Contract, even if such changes in law only come into force after the execution of the Contract.

- 1.3 The headings in these Conditions are for convenience only and shall be ignored in construing these Conditions and shall not affect their interpretation.

- 1.4 Words (including words defined in the Contract) importing the singular also include the plural and *vice versa* where the context requires. The words "written" and "in writing" include any means of visible reproduction.

2 Basis of the sale

- 2.1 The Seller shall sell and the Purchaser shall purchase the Supplies in accordance with the Contract.

- 2.2 In the event of any inconsistency between these Conditions and other documents forming part of the Contract, the following order of priority shall apply:

2.2.1 Any written agreement between the Parties where the Parties agree that any of the provisions in these Conditions should be superseded with an express reference to this Clause 2.2;

2.2.2 The Seller's quotation and documents (if any) incorporated by express reference on the face of the quotation;

2.2.3 These Conditions; and

2.2.4 The Order.

- 2.3 No variation to these Conditions shall be binding unless agreed in writing and signed by the authorised representatives of the Purchaser and the Seller. Any varying terms proposed by the Purchaser in its Order or any other document of the Purchaser shall not become part of the Contract.

3 Orders, Specifications and Cancellations

- 3.1 No Order shall be deemed to be accepted by the Seller unless confirmed in writing by the Seller's authorised representative.

- 3.2 The Purchaser shall be responsible to the Seller for ensuring the accuracy of the terms of any Order (including any applicable specification).

- 3.3 The quantity, quality and description of and any specification for the Supplies shall be those set out in the Seller's quotation (if accepted by the Purchaser) or the Order (if accepted by the Seller). The Seller reserves the right to make any changes in the specification of the Supplies which are required to conform to any applicable statutory requirements or, where the Supplies are to be supplied to the Purchaser's specification, which do not materially affect their quality or performance.

- 3.4 No Order which has been accepted by the Seller may be cancelled, varied or suspended by the Purchaser except with the agreement in writing of the Seller and on terms that the Purchaser shall indemnify the Seller in full against all loss (including loss of profit), costs (including the cost of all labour and materials used), damages, charges and expenses incurred by the Seller as a result of the cancellation, variation or suspension.

4 Price

- 4.1 The price of the Supplies shall be the price stated in the Order as accepted by the Seller or in the absence of such Order or acceptance, the written quotation of the Seller. Unless there is any variation as referred to in Clause 4.2, all prices quoted are valid for the period stated in the quotation (and if no period is stated, for up to 30 days only from the date of issuance of the quotation), after which time they may be altered by the Seller without giving notice to the Purchaser.

- 4.2 The quoted prices are based upon the scope of the Supplies referred to in the Seller's written quotation. In the event of any variation in, *inter alia*, the scope of Supplies ordered (including delivery dates from that of the Seller's offer) the Seller reserves the right to vary the prices quoted.

- 4.3 Except as otherwise agreed in writing between the Purchaser and the Seller, all prices are given by the Seller on an ex Works basis. Where the Seller agrees to deliver the Supplies otherwise than at the Seller's premises, the Purchaser shall be liable to pay the Seller's charges for transport, packaging and insurance.

- 4.4 For the provision of Services at the Site, the price shall exclude any incidental costs incurred by the Seller for the purposes and in the course of providing the Services i.e. travel costs, costs for transporting tools and equipment etc.

- 4.5 The price is exclusive of any applicable goods and services tax ("GST") or other government charge or duty, which the Purchaser shall be additionally liable to pay at the rate and in the manner from time to time prescribed by law. If exemption from taxes is claimed, the Purchaser must provide a certificate of exemption. Any additional costs incurred by the Seller pursuant to any Act of Parliament or any order or regulation made by any governmental body or department shall be paid by the Purchaser. The price shall be net of any withholding tax payable by the Purchaser, and in no event shall the Purchaser be entitled to withhold or deduct any such tax from the price.

- 4.6 The Seller shall be entitled to adjustment of the price (to be mutually agreed in writing) in the event of changes in law or engineering standards applicable to or affecting the Supplies after the execution of this Contract.

5 Terms of payment

- 5.1 After the acceptance of the Order, the Seller reserves the right to suspend performance or withhold delivery under an Order if the Seller in its reasonable opinion determines that the Purchaser's credit worthiness is unsatisfactory or that the Purchaser will not be able to pay the amounts due and payable under the Contract or if the Purchaser fails to make payment of any amount due and payable ("Suspension"). The Seller shall by written notice to the Purchaser inform the Purchaser of such Suspension.

- 5.2 The Purchaser shall pay the price of the Supplies within 30 days of the date of the Seller's invoice. Time is of the essence in respect of payment of the Contract price. Receipts for payment will be issued only upon request.
- 5.3 If the Purchaser fails to make any payment on the due date then, without prejudice to any other right or remedy available to the Seller, the Seller shall be entitled to:
- 5.3.1 suspend any further deliveries and/or Services to the Purchaser and terminate the same pursuant to Clause 10 (without being liable to the Purchaser for any losses);
- 5.3.2 at its sole discretion, apply any monies received from the Purchaser in relation to the Contract or any other contract or agreements between the Seller and the Purchaser, including but not limited to deposits or security payments, towards the payment of the relevant invoice; and
- 5.3.3 charge the Purchaser interest on the amount unpaid on a daily basis at the rate of one per cent (1%) per annum above the average Prime Lending Rate of the Development Bank of Singapore, the United Overseas Bank and the Oversea-Chinese Banking Corporation, from the due date until payment in full is made.
- 5.4 The Purchaser is not entitled to withhold from, set off against or otherwise reduce any payments due to the Seller unless agreed in writing by the Seller.
- 6 Delivery & Installation**
- 6.1 Unless otherwise agreed in writing, Supplies shall be delivered ex Works.
- 6.2 Partial delivery/performance of Supplies ordered is permissible. Where the Supplies are to be delivered/performed in instalments, each delivery/performance shall constitute a separate contract and failure by the Seller to deliver/perform any one or more of the instalments in accordance with the Contract or any claim by the Purchaser in respect of any one or more instalments shall not entitle the Purchaser to treat the whole Contract as having been repudiated.
- 6.3 Where in relation to the delivery/provision of Supplies, works need to be performed by the Seller at a Site, the Purchaser shall at its own costs and expense, take all necessary measures, to prepare the Site and ensure that the Site is suitable and ready for the commencement of the works or Services. The Purchaser shall ensure that the access roads to the Site are clear, the Seller's personnel are able to commence work immediately upon their arrival at the Site and further ensure that work proceeds in an uninterrupted manner. The Purchaser is responsible for all things related to the Site and shall supply the Seller with such labour, information, facilities, equipment and any other materials and tools which the Seller requires for the delivery/provision of the Supplies. This also includes information relating to the Site and all auxiliary services required by the Seller which are relevant for the purposes of the delivery/provision of the Supplies. Without limiting the foregoing, the Purchaser shall provide at its own expense and in a timely manner the following:
- 6.3.1 unrestricted access to the Site, including but not limited to entry permits and security passes;
- 6.3.2 such assistance as may be required by the Seller in relation to obtaining import, export and customs clearance for personal belongings and goods of the Seller and its subcontractors' personnel and of the equipment, tools and goods required for the works or Services to be performed at the Site;
- 6.3.3 assistance to the Seller and its subcontractor's personnel for the obtaining of visas, work and residential permits to the extent required for the carrying out of the works or Services at the Site as well as with regard to any permits required for leaving the country;
- 6.3.4 repatriation of the Seller's and its subcontractor's personnel in case of emergencies including but not limited to war, civil war, civil disturbance and epidemics;
- 6.3.5 the equipment and materials as may be required by the Seller which are necessary for the assembly, commissioning and the performance of the Services, such as scaffolds, lifting equipment and other devices as well as fuels and lubricants;
- 6.3.6 make available any information required concerning the location of concealed electric power, gas and water lines or of similar installations as well as the necessary structural data; and
- 6.3.7 all utilities including but not limited to energy, water, telecommunications services, heating and lighting;
- 6.3.8 suitable dry and lockable rooms of sufficient size adjacent to the Site for the storage of machine parts, apparatus, materials, tools, etc. and adequate working rooms for the Seller's and its subcontractors' personnel, including sanitary facilities as are appropriate in the specific circumstances;
- 6.3.9 all reasonable measures to protect the property of the Seller and its personnel at the Site;
- 6.3.10 protective clothing and protective devices needed due to particular conditions prevailing on the specific Site; and
- 6.3.11 documentation of existing systems (e.g. service manuals, operator guides), building descriptions and floor plans, programming devices, measurement, test resources and tools.
- 6.4 Where the Purchaser does not either provide or fulfil the above required obligations in Clause 6.3 to the satisfaction of the Seller, the Seller shall be entitled to carry out all necessary work at the costs and expenses of the Purchaser provided the Purchaser has been informed of the non-compliance and has failed to remedy this within the rectification period set out in such notice of non-compliance. The Purchaser shall not move any Goods, equipment or part thereof from the Site without the prior consent in writing of the Seller.
- 6.5 Subject to the Purchaser granting the Seller such reasonable access to the Site and the Site conditions being suitable for the performance of the works or Services in accordance with Clause 6.3, the Seller shall commence work at the Site on the scheduled date and/or deliver the Supplies on the scheduled date.
- 6.6 Where Supplies are supplied to the Purchaser and no assembly, installation, erection and commissioning is required:
- 6.6.1 The Purchaser is obliged to inspect the Supplies upon delivery and shall notify the Seller within one (1) week of receipt of the Supplies if there are any defects. Such notification of any defects shall be accompanied with relevant supporting evidence.
- 6.6.2 If the Purchaser fails to notify the Seller of the defects within one (1) week of the receipt of the Supplies, it will be deemed to have accepted such part of the Supplies.
- 6.7 Where installation, erection or commissioning is required:
- 6.7.1 The Purchaser shall accept the Supplies within two (2) weeks from the date that the Seller declares that the Supplies are ready for acceptance.
- 6.7.2 The Purchaser will be deemed to have accepted the Supplies if (a) the Supplies are put to use by the Purchaser for commercial purposes or for other purposes other than for testing; (b) it fails to respond to the Seller's declaration for acceptance; or (c) it fails to accept the Supplies within the two (2) weeks period without providing any written reasons or specific details of such refusal.
- 6.8 The Purchaser shall not be entitled to withhold acceptance for (a) defects which do not materially affect the usage of the Supplies; (b) minor deviations or deficiencies which do not materially affect the functioning of the Supplies; (c) defective installation or erection not carried out by the Seller; or (d) reasons which are not within the reasonable control of the Seller. As long as defects are capable of being remedied and Seller has not refused in writing to undertake the required remedial efforts, acceptance of the Goods and/or Services shall be deemed to have taken place upon completion of remedial efforts by Seller.
- 6.9 If the Goods or Services or any portion thereof is ready for delivery or performance and cannot be delivered or performed for reasons beyond Seller's control, acceptance shall be deemed to have taken place upon Seller's notification to Purchaser of readiness for delivery or performance.
- 6.10 Any costs and expenses related to the inspection and/or acceptance of the Supplies shall be borne by the Purchaser.
- 6.11 If the Seller fails to Deliver the Supplies by the Delivery Date (and in the case where Delivery is via instalment, by the last Delivery Date), and such failure is due to the sole fault of the Seller, the Seller is liable to pay to the Purchaser liquidated damages up to 0.5% of the portion of price (excluding any charge for transport, packaging or insurance) corresponding to that part of the Supplies which because of the delay could not be put to intended use for each consecutive seven-day period (inclusive of the seventh day) of delay, up to a maximum aggregate of 5% of such price. Such payment of liquidated damages shall be the Purchaser's sole and exclusive remedy for delay and shall be in lieu of any other rights the Purchaser may have against the Seller under the law.
- 6.12 The Delivery Date(s) may be extended, at the Seller's discretion and by written notification to the Purchaser, by such further periods as may reasonably reflect any delay which will or may be or has been caused by any of the following events:
- 6.12.1 Force Majeure, as provided in Clause 14;
- 6.12.2 Any variation or suspension of the Contract pursuant to Clauses 2.3 or 3.4;
- 6.12.3 Any suspension of the Contract pursuant to Clause 10;
- 6.12.4 Any breach of Contract by the Purchaser;
- 6.12.5 If the Supplies are not Delivered due to or in relation to the event described in Clause 6.13; or

- 6.12.6 Any other circumstances where the Seller is entitled under law to have an extension of time.
- 6.13 The Seller shall not be liable for any delays due to an act or omission of the Purchaser (including but not limited to any failure of the Purchaser to comply with any of its obligations under the Contract ("Purchaser's Default")). Where there is a Purchaser's Default, the Seller shall be entitled to claim such increase in costs incurred by the Seller as a result of the Purchaser's Default.
- 6.14 If the Purchaser fails to give the Seller adequate delivery instructions at the time stated for delivery then, without prejudice to any other right or remedy available to the Seller, the Seller may:
- 6.14.1 store the Supplies until actual delivery and charge the Purchaser for the reasonable costs (including insurance) of storage; or
- 6.14.2 sell the Supplies at the best price readily obtainable and (after deducting all reasonable storage and selling expenses) charge the Purchaser for any shortfall below the price under the Contract.
- 6.15 The Seller will give a credit note only for any Supplies or any part thereof that has been wrongly delivered/performed or over supplied.
- 6.16 Charges for all handling and transport as determined by the Seller are to be paid by the Purchaser except where such Supplies have been wrongly delivered or over supplied or where such charges have been incurred as a result of the Seller's performance of obligations during the warranty period under Clause 8.
- ## 7 Title and Risk
- 7.1 Unless otherwise agreed in writing between the Parties, risk of damage to or loss of the Supplies shall pass to the Purchaser as soon as they are Delivered.
- 7.2 Notwithstanding delivery and the passing of risk in the Supplies, or any other provision of these Conditions, title in the Supplies shall not pass to the Purchaser until the Seller has received in cash or cleared funds payment in full of the price of the Supplies and all other goods agreed to be sold by the Seller to the Purchaser for which payment is then due.
- 7.3 Until such time as title in the Supplies passes to the Purchaser, the Seller shall be entitled at any time to require the Purchaser to deliver up the Supplies to the Seller and, if the Purchaser fails to do so forthwith, to enter upon any premises of the Purchaser or any third party (whose cooperation the Purchaser hereby undertakes to procure) where the Supplies are stored and repossess the Supplies. Until such time as in accordance with this provision, the Supplies shall not be pledged or given as security or resold by the Purchaser and the Purchaser undertakes to store the Supplies in its premises separately from its own goods or those of any other person and in a manner which makes them readily identifiable as the Seller's Goods.
- 7.4 The Purchaser shall reimburse the Seller for any expenses and costs to the Seller in recovering any Supplies arising from any non-compliance by the Purchaser with the terms of Clause 7.3.
- ## 8 Warranties
- 8.1 The Seller warrants that it will perform the Services where required with reasonable care and skill and that the Supplies will correspond with their specification at the time of delivery and will be free from defects in material and workmanship under normal use and service for a period of twelve (12) months ("Warranty Period") from the date of Delivery of the Supplies.
- 8.2 If the Supplies are repaired or replaced during the Warranty Period, the period of Warranty for replacements shall run for six (6) months from the date of such repair or replacement but shall run at least until the expiry of the original warranty period as provided under Clause 8.1 and shall end at latest six (6) months after the end of the Warranty Period ("Extended Warranty Period").
- 8.3 The Seller shall be under no liability under Clause 8.1 and 8.2 whatsoever:-
- 8.3.1 in respect of any defect in the Supplies arising from any drawing, design or specification supplied by the Purchaser;
- 8.3.2 in respect of any defect arising from fair wear and tear, wilful damage, negligence, abnormal working conditions, failure to follow the Seller's instructions (whether oral or in writing), misuse or alteration or repair of the Supplies without the Seller's approval or improper or inadequate maintenance by the Purchaser;
- 8.3.3 in respect of minor deviations from the drawings, design or specifications supplied by the Seller, insignificant deviations from the agreed quality or minor impairment of usability which do not materially affect the commercial use of the Supplies;
- 8.3.4 if the Supplies have been used in a manner or under a circumstance or for a purpose not reasonably to be inferred by the Seller or disclosed to the Seller prior to making the Contract;
- 8.3.5 if the total price for the Supplies has not been paid by the due date for payment;
- 8.3.6 for any delay in the delivery or installation of the Supplies if such delay arises or results from variation of the Contract pursuant to Clause 2.3 or 3.4;
- 8.3.7 if the Purchaser permits persons other than the authorised representative of the Seller to effect any replacement of parts, maintenance adjustments or repairs to the Supplies; or
- 8.3.8 where the Supplies are Software, for non-reproducible software errors.
- 8.4 Save for the warranties specified in Clause 8.1, all other warranties, conditions or other terms implied by statute or common law are excluded to the fullest extent permitted by law. The Seller specifically disclaims the implied warranties of merchantability and fitness for a particular purpose.
- 8.5 During the Warranty Period and Extended Warranty Period (as the case may be), any claim by the Purchaser which is based on any defect in the quality or condition of the Supplies or their failure to correspond with specification shall (whether or not delivery is refused by the Purchaser) be notified to the Seller within the relevant period referred to in Clause 6.6 and 6.7 or (where the defect or failure was not apparent on reasonable inspection) within 7 days after discovery of the alleged defect or failure, failing which the Seller shall not be liable under the warranty provided in Clause 8.1 herein.
- 8.6 Where any valid claim in respect of any of the Supplies which is based on any defect in the quality or condition of the Supplies or their failure to meet specification is notified to the Seller in accordance with these Conditions, the Seller's sole liability shall be to repair the Supplies or replace the Supplies (or the part in question) free of charge at the Seller's sole discretion, or refund to the Purchaser the price of the defective Supplies (or a proportionate part of the price).
- 8.7 The benefit of this Warranty shall apply only to the Purchaser.
- 8.8 Clause 8 sets out the Purchaser's sole and exclusive remedy for all warranty claims during the Warranty Period and Extended Warranty Period.
- ## 9 Intellectual Property Rights
- 9.1 If any claim, demand, action or proceeding is brought against the Purchaser for alleged infringement of any registered design or trade mark or trade name or copyright or letters patent (the specifications of which are published prior to the date of the Order) by the Supplies or any part thereof or any allegation of such infringement is made, and provided that the Purchaser does not concede the existence of an infringement but gives the Seller immediate notice in writing of any such allegations or infringement or of the institution of any such action or proceeding and permits the Seller to answer the allegation and to conduct all negotiations in respect of such allegation and to defend the action or proceeding, and also provided that the Purchaser gives the Seller (at the Seller's costs and expense) all information assistance and authority required for those purposes and does not by any act (including any admission or acknowledgment) or omission prejudice the conduct of such defence, then:
- 9.1.1 The Seller will at its own election either effect any settlement or compromise which it deems reasonable or at its own costs and expense defend any such action or proceeding and if the Supplies or any part thereof is in such action or proceeding held to constitute infringement and is the subject of an injunction restraining its use or any order providing for its delivery up or destruction the Seller shall at its own election and costs and expense either:
- (i) procure the Purchaser the right to retain and continue to use the Supplies or part thereof; or
 - (ii) modify the Supplies or any part thereof so that it becomes non-infringing; or
 - (iii) replace the Supplies or any part thereof with one that is non-infringing.
- 9.2 The Seller shall not be under any of the obligations specified in Clause 9.1 hereof in any of the following events:
- 9.2.1 Any infringement or allegation thereof which is due to the Seller having followed a design or instruction furnished by the Purchaser or based upon the use of the Supplies in a manner or for a purpose not reasonably to be inferred by the Seller or disclosed to the Seller prior to making the Contract or in combination with other goods or devices or services not made or supplied by the Seller; or
- 9.2.2 The Purchaser entering into any compromise or settlement in respect of such action or proceeding without the Seller's prior consent in writing; or
- 9.2.3 Modifications of the Supplies by the Purchaser or a third party.
- 9.3 If the Purchaser stops using the Supplies in order to reduce the damage or for any other good reason, it shall be obliged to point out to the third party that no acknowledgement of the alleged infringement may be inferred from the fact that the use has been discontinued.

- 9.4 The Seller's obligation to reimburse the Purchaser for any claims resulting from the infringement of any intellectual property rights shall expire three (3) years from the start of the Warranty Period.
- 9.5 Save as provided in Clause 9, the Seller shall not be liable for any damage or losses incurred by the Purchaser arising from the use or non-use of any infringing Supplies or any part thereof.
- 9.6 If the Supplies are to be manufactured or any process is to be applied to the Supplies by the Seller in accordance with a specification submitted by the Purchaser, the Purchaser represents and warrants that any design or instructions furnished or given by it shall not be such as will cause the Seller to infringe any registered designs or trade mark or trade name or copyright or letters patent in the performance of the Contract and shall indemnify the Seller against all loss, damages, costs and expenses awarded against or incurred by the Seller in connection with or paid or agreed to be paid by the Seller in settlement of any claim for infringement of any patent, copyright, design, trade mark or other industrial or intellectual property rights of any other person which results from the Seller's use of the Purchaser's specification.
- 9.7 The patent and design rights held by the Seller shall remain the absolute property of the Seller and such designs and drawings, equipment or any part thereof shall not be reproduced or disclosed or allowed to be reproduced or disclosed by the Purchaser without the Seller's consent in writing.
- 9.8 Without prejudice to Clause 9.7, rights, title and interest in and to any intellectual property in the Supplies shall remain the property of the Seller or its licensors at all times.
- 10 Termination and/or Suspension of Supplies**
- 10.1 In the event that:
- 10.1.1 the Purchaser is in breach of the Contract; or
- 10.1.2 the Purchaser makes any voluntary arrangement with its creditors or (being an individual or firm) becomes bankrupt or (being a company) goes into liquidation (otherwise than for the purposes of amalgamation or reconstruction) or has an order made or resolution passed for such winding-up or shall otherwise become insolvent or make such proposal, assignment or arrangement for the benefit of its creditors or have a receiver or manager appointed over its affairs or have an application made to court for the appointment of a judicial manager or be placed under a judicial management order; or
- 10.1.3 an encumbrancer takes possession of, or a receiver is appointed over, any of the property or assets of the Purchaser; or
- 10.1.4 the Purchaser ceases or threatens to cease to carry on business; or
- 10.1.5 there is a change in control of the Purchaser which in the reasonable opinion of the Seller adversely affects the position, rights or interests of the Seller. (For the purpose of this sub-clause, "control" means the ability to direct the affairs of another whether by virtue of contract, ownership of shares, or otherwise howsoever); or
- 10.1.6 in the reasonable opinion of the Seller, there occurs a material change in the financial position of the Purchaser which is likely to affect the Purchaser's ability to perform its obligations under the Contract; or
- 10.1.7 The Seller reasonably apprehends that any of the events mentioned above is about to occur in relation to the Purchaser and notifies the Purchaser accordingly,
- the Seller shall be entitled to, by issuance of a written notification, (i) terminate the Contract forthwith on the occurrence of each of the events in Clauses 10.1.2, 10.1.3, 10.1.4 or 10.1.5 or suspend any further deliveries, production or commissioning works or the Services under the Contract without any liability to the Purchaser on the occurrence of each of the events in Clauses 10.1.1, 10.1.6 and 10.1.7, and upon the issuance of Seller's written notification of such termination or suspension, (ii) if the Supplies have been delivered or completed but not paid for, the price shall become immediately due and payable notwithstanding any previous agreement or arrangement to the contrary, and (iii) the Seller shall be entitled to retain any security given or monies paid by the Purchaser and apply the said security or monies against the assessed losses and damages, if any, suffered by the Seller, or if there is no such security or monies paid by the Purchaser, to recover the same otherwise.
- 10.2 Upon the issuance by the Seller of a written notification giving notice of the suspension of any further deliveries, production or commissioning works or the Services under the Contract pursuant to Clause 10.1, the Purchaser shall:
- 10.2.1 if the suspension is due to or arises from a breach by the Purchaser of the Contract under Clause 10.1.1, remedy such breach within fourteen (14) days of the written notification of suspension, failing which the Seller shall be entitled to terminate the Contract forthwith by issuance of a written notice to the Purchaser; or
- 10.2.2 if the suspension is due to or arises from a material change in the Purchaser's financial position in accordance with Clause 10.1.6, provide the Seller with a bank guarantee or such other security ("Security") to the Seller's reasonable satisfaction within thirty (30) days of the written notification of suspension, failing which the Seller shall be entitled to continue the suspension until the Seller is provided with such Security or to terminate the Contract forthwith by issuance of a written notice to the Purchaser; or
- 10.2.3 if the suspension is due to or arises from a Clause 10.1.7 event, provide the Seller with proof to the Seller's reasonable satisfaction, within thirty (30) days of the written notification of suspension, that the relevant event, the occurrence of which the Seller reasonably apprehends, is not occurring and will not occur, failing which the Seller shall be entitled to continue the suspension until the Seller is provided with such proof or to terminate the Contract forthwith by issuance of a written notice to the Purchaser.
- In the event the Contract is terminated pursuant to any of Clauses 10.2.1 to 10.2.3, the Seller shall, in addition and without prejudice to the remedies in Clause 10.1, be entitled to recover all losses and damages from the Purchaser.
- 10.3 The losses and damages in Clauses 10.1(iii) and 10.2 shall include but are not limited to all costs, expenses and liabilities reasonably incurred by the Seller in respect of Goods that are partially completed or Services not fully performed as well as any costs, expenses and liabilities in expectation of the completion of the Supplies. Termination of the Contract by the Seller shall not discharge the Purchaser from any existing obligation accrued due on or prior to the date of termination.
- 10.4 The rights and remedies granted to the Seller pursuant to the Contract are in addition to, and shall not limit or affect, any other rights or remedies available at law or in equity.
- 11 Compliance with Export Control Regulations**
- 11.1 If Purchaser transfers goods (hardware and/ or software and/ or technology as well as corresponding documentation, regardless of the mode of provision) delivered by Seller or works and services (including all kinds of technical support) performed by Seller to a third party, Purchaser shall comply with all applicable national and international (re-) export control regulations. In any event of such transfer of goods, works and services, Seller shall comply with the (re-) export control regulations of the Federal Republic of Germany, of the European Union and of the United States of America.
- 11.2 Prior to any transfer of goods, works and services provided by Seller to a third party, Purchaser shall in particular check and guarantee by appropriate measures that:
- (i) There will be no infringement of an embargo imposed by the European Union, by the United States of America and/ or by the United Nations by such transfer, by brokering of contracts concerning those goods, works and services or by provision of other economic resources in connection with those goods, works and services, also considering the limitations of domestic business and prohibitions of by-passing those embargoes;
- (ii) Such goods, works and services are not intended for use in connection with armaments, nuclear technology or weapons, if and to the extent such use is subject to prohibition or authorization, unless required authorization is provided;
- (iii) The regulations of all applicable Sanctioned Party Lists of the European Union and the United States of America concerning the trading with entities, persons and organizations listed therein are considered.
- 11.3 If required to enable authorities or Seller to conduct export control checks, Purchaser, upon request by Seller, shall promptly provide Seller with all information pertaining to the particular end customer, the particular destination and the particular intended use of goods, works and services provided by Seller, as well as any export control restrictions existing.
- 11.4 Purchaser shall indemnify and hold harmless the Seller from and against any claim, proceeding, action, fine, loss, cost and damages arising out of or relating to any noncompliance with export control regulations by Purchaser, and Purchaser shall compensate Seller for all losses and expenses resulting thereof.
- 11.5 Seller shall not be obligated to fulfill the Contract if such fulfillment is prevented by any impediments arising out of national or international foreign trade or customs requirements or any embargoes or other sanctions.
- 12 Software**
- 12.1 Where applicable, the Seller grants to the Purchaser a non-exclusive, non-transferable licence to use the Software delivered with the Supplies for the purposes stated in the manual of the Supplies during the validity of the Contract subject to the provisions of this Clause 12.

- 12.2 The Purchaser acknowledges that the Software contains valuable confidential and proprietary information and trade secrets of the Seller or its licensors, and undertakes that it (including its agents, employees and/or servants) shall not, without the Seller's prior consent in writing, disclose the details of the Software to third parties.
- 12.3 All rights, title and interest in the Software, including revisions and updated versions, shall remain the property of the Seller or its licensors. All rights to the Software not expressly granted in the Contract shall remain reserved.
- 12.4 In the event the Software contains third party components which the Seller has licensed under generally used "open source" licence terms, the terms of the Contract shall apply to those components to the extent that they do not conflict with the "open source" licence terms. If necessary, the Purchaser agrees to sign a licence agreement with the licensor of the Software.
- 12.5 The Purchaser is not allowed to make any back-up copies of the Software without the Seller's prior consent in writing.
- 12.6 The use of the Software on hardware other than the agreed equipment requires the Seller's consent in writing.
- 12.7 Unless otherwise agreed, the Software shall be provided in machine-readable form (object code) only.
- 12.8 The Seller may terminate the Purchaser's licence upon notice of breach of these licence terms. The Purchaser must destroy all copies of the Software immediately upon notice of termination. The Purchaser shall not disassemble or otherwise modify the Software without express agreement in writing from the Seller.
- ### 13 Limitation of Liability
- 13.1 Notwithstanding anything to the contrary in the Contract and except to the extent required by law, the total liability of the Seller, the Seller's personnel, the Seller's affiliates as well as the Seller's sub-contractors for any act or omission, whether in contract, tort (including negligence or strict liability), by way of indemnities or any other legal or equitable theory shall not exceed:
- (i) (in the case of Supplies where the contract price is calculated on an annual basis) 20% of the contract price of the Supplies for the preceding 12 calendar months' (calculated from the date of the breach); or
 - (ii) (in all other instances) 20% of the price of the Supplies under the Contract.
- 13.2 The Seller shall not be liable to the Purchaser for any loss of profit (actual or anticipated), loss of use, loss of production (including loss of hydrocarbons), loss of contracts, loss of opportunities, loss of revenue, cost of capital, costs of replacement, loss of goodwill, loss of reputation, loss of information or data, loss from any third party contracts, loss due to business interruption, loss of interest, loss of power, cost of purchased or replacement power, contractual claims from third parties or any indirect, incidental, special or consequential losses or damages arising from or in connection with its performance or non-performance under the Contract and whether based upon contract, tort, or any other legal theory. Clause 13.2 shall apply to the benefit of the Seller's personnel, the Seller's affiliates and the Seller's sub-contractors.
- 13.3 If the Purchaser is not or shall not be the sole end user and ultimate owner of the Goods or beneficiary of the Services or is procuring them for any joint venture, the Purchaser shall include a clause in its contract with the end user, ultimate owner or joint venture participants such that the Seller is given the benefit of the indemnities, exclusions and limitations of liability in this Contract by all such users, owners or participants (which shall apply as if the user, owner or participant were the Purchaser) and shall indemnify the Seller against claims by them to the extent that the Seller would not be liable to the Purchaser under this Contract if the claim had been made by the Purchaser.
- ### 14 Force Majeure
- 14.1 The Seller shall not be liable to the Purchaser or be deemed to be in breach of the Contract by reason of any delay in performing, or any failure to perform, any of the Seller's obligations in relation to the Supplies, if the delay or failure was due to force majeure. For the purposes of this clause, force majeure shall mean any unforeseen event beyond the reasonable control of the Seller such as, but not limited to, any act of God, act of government or any authorities, non-issuance of licences, hostilities between nations, war, riot, civil commotions, civil war, insurrection, blockades, import or export regulations or embargoes, rainstorms, national emergency, earthquake, fires, explosion, flooding, hurricane or other exceptional weather conditions or natural disaster, acts of terrorism, accidents, sabotages, strikes, shortages in material or supply, infectious diseases, epidemics, as well as travel restrictions or travel warnings due to any such events. If any delay in performing, or any failure to perform the Contract is caused by the delay of a subcontractor of the Seller, and is beyond the control and without the fault or negligence of the Seller, the Seller shall incur no liability for such delay.
- 14.2 Where there is force majeure, the Seller is entitled to an extension of the Delivery Date(s) as provided in Clause 6.
- 14.3 Without prejudice to the other provisions of this Contract, where the force majeure continues for more than six (6) months, the Seller shall have the right to terminate the Contract. In such a situation, the Seller shall be reimbursed for Supplies already performed/delivered, cost of materials or goods reasonably ordered, any other expenditure reasonably incurred in the expectation of completing the Supplies as well as the reasonable cost for removal of the Seller's equipment and demobilization of personnel.
- ### 15 General
- 15.1 The Seller is a member of the group of companies ("the Siemens group of companies") whose holding company is Siemens Aktiengesellschaft ("SAG"), and accordingly the Seller may perform any of its obligations or exercise any of its rights hereunder by itself or through any other member of this group, provided that any act or omission of any such other member shall be deemed to be the act or omission of the Seller.
- 15.2 The Purchaser shall not assign any of its rights or obligations under the Contract without the prior written consent of the Seller, such consent to be signed by its authorised representatives. Any attempted delegation or assignment shall be void. The Seller may attach conditions to the giving of its consent.
- 15.3 The Seller may assign any of its rights (including receivables) under the Contract without the prior written consent of the Purchaser.
- 15.4 The Seller may sub-contract the performance of the Contract or any part of the Contract without obtaining the prior consent of the Purchaser.
- 15.5 Any shipment terms quoted in relation to the delivery of the Supplies shall be in accordance to Incoterms 2010 or its latest version as from time to time modified supplemented or revised.
- 15.6 The Purchaser shall comply with all relevant statutes, rules and regulations and by-laws affecting its obligations and the performance of the Contract and shall obtain at its own costs and expenses all necessary permits and licences and shall furnish to the Seller, upon request, information or documentation of the Purchaser's compliance, as well as to any other information or documentation required to enable the Seller to comply with any laws, rules, regulations and requirements applicable to its performance of the Contract.
- 15.7 Any notice required or permitted to be given by either Party to the other under these Conditions shall be in writing and signed by the authorised representatives of the Party addressed to that other Party at its registered office or principal place of business or such other address as may at the relevant time have been notified to the Party giving the notice. Notices may be delivered by hand, or by prepaid registered post or by facsimile and shall be deemed to have been served:
- (i) if by hand, at time of delivery;
 - (ii) if by prepaid registered post, 3 working days after posting; or
 - (iii) if by facsimile, on the date printed on the facsimile transmission report produced by the sender's machine.
- 15.8 No waiver by the Seller of any breach of the Contract by the Purchaser shall be considered as a waiver of any subsequent breach of the same or any other provision. If the Seller delays, targets or chooses not to enforce its right under the Contract, this shall not affect its right to do so at a later date.
- 15.9 If any provision of these Conditions is held by any competent authority to be invalid or unenforceable in whole or in part such provision shall be construed, limited or if necessary, severed to the extent necessary to eliminate such invalidity or unenforceability and the validity of the other provisions of these Conditions and the remainder of the provision in question shall not be affected but shall remain in full force and effect.
- 15.10 No terms shall survive the expiry or termination of the Contract unless expressly provided.
- 15.11 The Contract is the entire agreement between the Parties and may not be changed unless agreed in writing by properly authorised representatives of both Parties. The Contract shall supersede any other express or implied, written or oral terms, arrangements, customs or practices.
- 15.12 Save for the Siemens group of companies, the Parties do not intend that any term of the Contract should be enforceable, by virtue of the Contracts (Rights of Third Parties) Act (Chapter 53B of Singapore) or otherwise, by any person who is not a party to the Contract.

16 Applicable Law and Dispute Resolution

- 16.1 The Contract (including these Conditions) shall be governed by and construed in accordance with the laws of Singapore. The application of the United Nations Convention on Contracts for the International Sale of Goods of April 11, 1980 shall be excluded.
- 16.2 The Parties will use their best efforts to negotiate in good faith and settle any dispute that may arise out of or relate to this Contract or any breach of it. If any such dispute cannot be settled amicably through ordinary negotiations between representatives of the Parties, the dispute shall be referred to the management of each Party who will meet in good faith in order to try and resolve the dispute.
- 16.3 All negotiations connected with the dispute will be conducted in complete confidence and the Parties undertake not to divulge details of such negotiations except to their professional advisers who will also be subject to such confidentiality obligations and such negotiations shall be without prejudice to the rights of the Parties in any future proceedings.
- 16.4 In the event any such dispute is unresolved after thirty (30) days of the commencement of such negotiations referred to in Clause 16.2, such disputes arising out of or in connection with the Contract, including any question regarding its existence, validity or termination, shall be referred to and finally resolved by arbitration administered by the Singapore International Arbitration Centre ("SIAC") in accordance with the Arbitration Rules of the Singapore International Arbitration Centre ("SIAC Rules") for the time being in force, which rules are deemed to be incorporated by reference in this Clause. For disputes where the total quantum in dispute (including all counterclaims) is less than S\$5 million, the tribunal shall consist of one arbitrator to be appointed by the President of the Court of Arbitration of SIAC. For all other disputes, the tribunal shall consist of three arbitrators to be appointed by the President of the Court of Arbitration of SIAC. The seat of arbitration shall be Singapore. The language to be used in the arbitration proceedings shall be English.

17 Environment, Health & Safety

- 17.1 Seller and Purchaser shall comply with all health and safety laws and regulations in force at the work area, which in particular includes without limitation the ambient air of such area on any site where: (i) the Goods need to be installed or (ii) the Services need to be provided (hereinafter "Work Area") as well as with the Workplace Safety and Health Act (Chapter 354A of Singapore). Where applicable, Purchaser shall provide to Seller the Work Area at the agreed point in time and shall ensure that the Work Area and the Purchaser's facility where the work is to be performed ("Site") complies at such time with all aforementioned laws, regulations and requirements, and shall not during the performance of this Contract cause any risks not in compliance with the aforementioned laws, regulations and requirements.
- 17.2 Purchaser shall inform Seller in writing prior to the execution of any work about possible health or safety risks which may arise out of Purchaser's plant and equipment or may exist on Site, including but not limited to hazardous materials, which may exist in addition to any hazardous materials already specifically identified in the Contract to exist or the existence of which is obvious for the Seller. If agreed by the Parties during project execution, the Seller shall provide reasonable support to the Purchaser at the Purchaser's expense in the assessment of the existence of any such health or safety risks and what protective and preventive measures should be taken.
- 17.3 Seller's work does not comprise the use of special protective and preventive measures against any health or safety risks for Seller's and its subcontractors' personnel in or arising out of Purchaser's plant or equipment or existing on Site to the extent such measures go beyond what is normal and usual for the type of Services to be performed by Seller. However, Seller shall be responsible for the clearance of any health or safety risk in the Work Area to the extent this has been created due to his fault. In the given case of health or safety risks for which Seller is not responsible as defined in the two preceding sentences above (including but not limited to the exposure to artificial mineral fibers or asbestos concentrations in the Work Area or an area used for access to the Work Area are greater than 1,000 particles / m3 of ambient air) ("Purchaser Health or Safety Risks") the Parties shall reach agreement in good faith on how to proceed. However, Seller shall not be obligated to carry out any work at all if the asbestos concentration exceeds 100,000 particles / m3 ambient air.
- 17.4 In case of reasonable doubt regarding the existence of Purchaser Health or Safety Risks the Purchaser shall provide proof of their non-existence by means of the expertise of an independent, internationally operating, specialized and qualified company or institute agreed upon by the Parties. The expertise shall take into account all circumstances, including, but not limited to the environment of the Site and shall adhere to the Siemens "Annex Working under Hazardous Conditions - Asbestos Guideline", which shall be provided to Purchaser upon request. Nothing in this Contract shall in any way be construed to limit Seller, in its sole discretion, from taking any samples or measurements of such areas, though Seller shall have no obligation to do so.

- 17.5 In the event there are Purchaser Health or Safety Risks or there exists any reasonable doubt that the same exist or may be created by or during the work or the Purchaser has failed to comply with any of its obligations set out in the preceding paragraphs, then, without limiting its other rights and remedies, Seller shall be entitled to suspend its work until: (i) the respective Purchaser Health or Safety Risk has been permanently eliminated or (ii) the addressed reasonable doubt has been proven to be unsubstantiated or (iii) agreed protective and preventive measures have been taken.
- 17.6 Purchaser shall dispose of all waste created on its Site in a timely manner and in accordance with all applicable laws. The Purchaser shall also be responsible for the timely and lawful disposal of its equipment or parts thereof, which are or become hazardous waste. In all cases Purchaser shall be responsible for decontamination and/or disposal of any radioactive materials or waste.
- 17.7 Purchaser shall indemnify and hold Seller harmless from and against any claims, losses or damages any employee of Seller or its subcontractors or any other third party may demand from Seller and arising out of the realization of any Purchaser Health or Safety Risks or the disposal of any waste that the Purchaser is responsible for.